

WORKSHOP

Details

Introduction

The workshop will be conducted over **four weeks**, with sessions held every **Tuesday and Thursday**, for a total of eight sessions. In addition, two complimentary sessions will be conducted: one covering Powertrain and Embedded/Electrical fundamentals, and another covering Mechanical fundamentals. The timings for these sessions will be communicated separately.

The first session will be on the 1st of October 2026 (Thursday). The presession talk is scheduled for the 29th September 2026. Following this we'll have sessions every Tuesdays and Thursdays.

The workshop will focus on hands-on learning, supported by handbooks, guides, and other resources to help participants understand the concepts and apply them effectively.

Note: If any topic requires additional time beyond the allocated session, supplementary online sessions or alternative arrangements will be made to ensure that the planned content is adequately covered.

The Syllabus

- ROS
- ML
- Computer Vision
- Agentic AI
- System design

Detailed content covered in each topics

ROS

- a. Core Primitives: Understanding Nodes (individual programs), Topics (data channels), and Messages (data packets), Basic ROS Commands
- b. The Publisher-Subscriber Model: Explaining communication using an onboard analogy (e.g., a wheel speed sensor publishes data; a dashboard display subscribes to it).
- c. Mini Project -

System Design

- a. Subsystem Thinking: Breaking a complex ATV down into inputs, processing units, and outputs across mechanical, electrical, and software domains.
- b. Block Diagramming: Learning standard conventions for mapping data flow, power distribution, and physical linkages cleanly.
- c. Mini Project -

ML & Computer Vision

- a. Understanding pixels, resolutions, and channels.
- b. Image manipulation basics: resizing, cropping, and applying Gaussian blur to reduce camera noise.
- c. Color & Contour Tracking
- d. Core Paradigms: Supervised learning (learning from labeled vehicle logs) versus unsupervised learning (grouping driving styles).
- e. Linear Regression Demystified: Predicting continuous values, such as fuel/battery consumption based on throttle percentage.
- f. Mini Project -

Agentic AI

- a. Introduction to Generative AI
- b. AI Agents & Agentic Systems
- c. Prompting Fundamentals
- d. Tools & Decision Making
- e. Engineering Applications of AI
- f. Mini Project -

Bonus sessions

Powertrain

- a. Basic Electrical Concepts
- b. Digital & Analog Signals
- c. Microcontrollers
- d. Embedded Programming Fundamentals
- e. GPIO & Peripheral Interface

Mechanical fundamentals

- f. Vehicle Architecture
- g. Chassis & Components
- h. Steering & Suspension
- i. Braking & Drivetrain
- j. Sensor Mounting
- k. Basics of Vehicle Dynamics

Timeline

- Every session will be conducted between 5:10 to 6:50 pm on the days specified.
- The timing for the bonus sessions will be informed later.

Week 0

Pre-Workshop | Online Beginner Session

- An online session will be conducted before the commencement of the workshop.
- The session will cover the prerequisites, requirements, and necessary setup for the workshop.

- Participants will be given a basic overview of the workshop and the concepts that will be covered.

Week 1

Tuesday: System Design

Thursday:

Week 2

Tuesday: Computer Vision

Thursday: ML

Saturday | Bonus Session I

Powertrain & Embedded/Electrical

Week 3

Tuesday: ML

Thursday: ROS

Saturday | Bonus Session II

Mechanical Fundamentals

Week 4

Tuesday: Agentic AI – I

Thursday: Agentic AI – II

Feel free to reach out to us with any questions or doubts throughout the workshop. We encourage you to make the most of the sessions, resources, and hands-on opportunities provided. If you encounter any difficulties or have suggestions for improving the workshop, we would be happy to hear from you.

**The syllabus, content covered, schedule, and other workshop details are subject to change with or without prior notice.*